Flood forecasting of lower Teesta River basin using HEC-HMS

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Abstract:

The Teesta River flows from the beautiful Himalayan glaciers in the Indian state of Sikkim. Its primary tributaries, such as the Rangeet and Rangpo rivers, are likewise Himalayan in origin. The Teesta River flows through the Indian states of Sikkim and West Bengal before joining the Brahmaputra River in Bangladesh. The river is roughly 315 kilometers long in India and 172 kilometers long in Bangladesh. In the monsoon, heavy rainfall on the upper basin of Teesta brings much water which causes flood inside Bangladesh. This flood affects on livings and damages economy of the study area, Rangpur Division. This study has tried to build a flood forecasting model using HEC-HMS which will give early warning to flood by calculating peak discharge from the satellite data. In this study, authors calibrated the precipitation data of 2014 with in situ data. From the result, the percent bias is 5.16%. Later data from the year 2015 and 2016 have been validated with in situ data and the percent bias was 3.70% and 5.71% respectively. Thus the model can be used for verifying the future data and can forecast the flood which will help to mitigate the effects of flood in the lower Teesta basin located inside Bangladesh.

Introduction:

HEC-HMS is a mathematical watershed model that contains several methods with which to simulate surface runoff and river/reservoir flow in river basins. The hydrologic model, together with flood damage computations (also included in the model), provides a basis for the evaluation of flood control projects.

The Lower Basin of the Teesta River, nestled in the northeastern region of the Indian subcontinent, is no stranger to the devastating impacts of flooding. This picturesque landscape, known for its breathtaking beauty and vital agricultural activity, is paradoxically vulnerable to the destructive forces of nature. The Teesta River, originating from the Tibetan Plateau and flowing through India and Bangladesh, plays a pivotal role in shaping the livelihoods of millions. However, the unpredictability of its flow patterns, exacerbated by climate change and rapid urbanization, has transformed the serene Teesta into a formidable force that frequently breaches its banks, inundating homes, farmlands, and critical infrastructure. These floods primarily result from a combination of factors, including heavy monsoon rains and the release of water from dams and reservoirs upstream. The Teesta River and its tributaries, which flow through this region, play a significant role in shaping the flood dynamics in the area.

For this project, we studied two papers. One of this, “Application of HEC-HMS for flood forecasting in Misai and Wan’an catchments in China”. The paper worked for two catchments: the Misai and Wan’an Catchments.

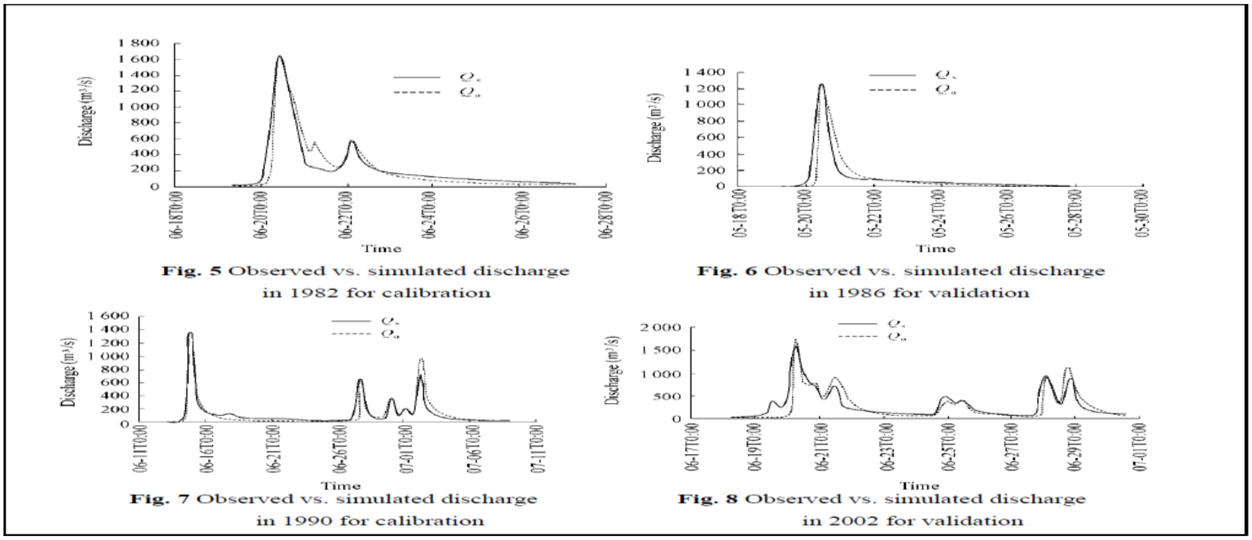


Fig. 5 Observed vs. simulated discharge in 1982 for calibration

Fig. 6 Observed vs. simulated discharge in 1986 for validation

Fig. 7 Observed vs. simulated discharge in 1990 for calibration

Fig. 8 Observed vs. simulated discharge in 2002 for validation

Figure 1*:Comparison between simulated and observed discharge*

Observed Discharge Data and Simulated Discharge Data calibration and Validation curves. It can be seen in the above graphs that the simulated and observed peak discharges occurred on the same day, and their maximum time difference was one hour, which is acceptable for flood forecasting.

Another one is Flash flood risk assessment for the upper Teesta river basin: using the hydrological modeling system (HEC-HMS) software (Subhra Prakash Mandal, Abhisek Chakrabarty). They mainly construct a unit of hydrograph for an excess rainfall event by estimating the stream flow response at the outlet of each sub-watershed and simulation of the extent of inundation of the river banks.

Study Area:

The Study area of this project is the lower basin of Teesta River, the northeastern region of India and extending into BangladeshThe Teesta River lower basin extends from Sikkim in India in the eastern Himalayas, through West Bengal to the northern Rangpur division in Bangladesh. The river joins the Brahmaputra in Bangladesh before it flows into the Bay of Bengal after meeting with the Ganges and the Meghna.

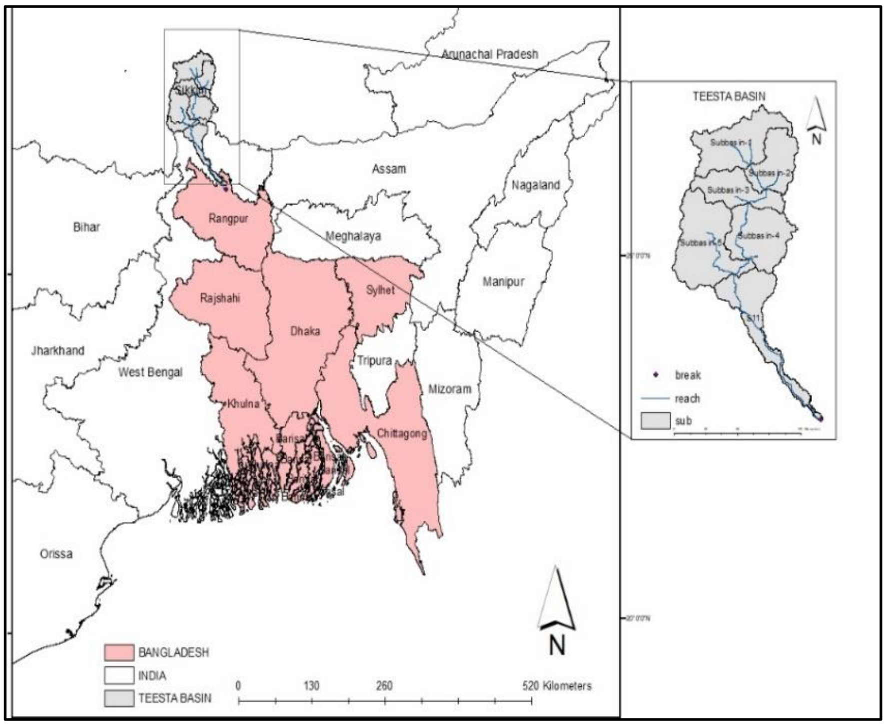


__Figure 2:__ Study Area Map (Lower Teesta Basin)

Methodology

Data collection

DATA	PERIOD	SOURCE
DISCHARGE	2014-1016	BWDB Dalia Station (SW291.5R)
PRECIPITATION	2014-2016	Copernicus (ERA5)
DEM	-	Open Topography

Table:1

We downloaded USGS 1 arc-second Digital Elevation Model from Open Topography. DEM resolution is 30m. We collected precipitation data from cds.climate.copernicus.eu website using ERA5 satellite. we downloaded hourly data for our study area for the year 2014 to year 2016. Our downloaded data is in .netCDF format. To convert those data to CSV format we used Python programming language. At first, we ran the Python code in Jupiter Notebook and the code provided us two latitudes and six longitudes. Among those latitude and longitude, we chose the latitude and longitude as per our sub-basin coordinates. And we get two different pairs of latitudes and longitudes. We assign those latitudes and longitudes under two precipitation gauges. Sub-basin 1 to sub-basin 5 falls under precipitation gauge-1 and sub-basin falls under precipitation gauge-2. For our calibration and validation, we collected discharge data from year 2014 to year 2016 from BWDB Dalia station (SW291.5R). We used discharge data for 2014 for calibration and 2015, 2016 for our validation purposes.

Model Setup:

To create our model, a new terrain and a basin model were created. Then our dem was imported into that terrain. A coordinate system was set for our terrain. WGS 1984_45N was used.

Then preprocess sink, preprocess drainage, identifying streams, breakpoint addition, and delineating elements were done respectively. In identifying streams we needed to enter the area. An area of 500 km^2 was set and in breakpoint manager Dalia station point shape file was added and then delineated. After delineating,11 sub-basins and 5 reaches (reach 1 to 5 were created. Some of them were merged and finally 6 sub-basins were obtained (Sub-basin 1 to Sub-basin 5 and Sub-basin 11). Then the downloaded precipitation data was set in two precipitation gages, sub-basin 1-5 under gage-1 and sub-basin 11 under gage-2.

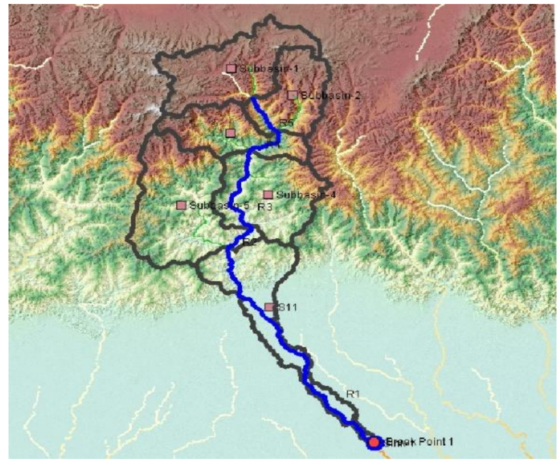
The methods we used in the model are as follows:

Loss method: SCS Curve Number

Transform method: Snyder's Unit Hydrograph.

Baseflow method: Recession

Routing method: Muskingum



First, some reasonable values for the parameters in these methods were assumed. Then the simulation was run.

<u>Calibration and Validation:</u>

Calibration and validation were done for the model to get the most accurate results like observed data. Our model is calibrated for the year 2014. The observed data is daily discharge. For our model, there are several parameters to calibrate.

Routing method: Muskingum

Initially the value of "Muskingum K" is taken at 80 and the value of "Muskingum X" is taken at 0.5. After calibration, the "Muskingum K" is taken 120 for reach "R5", 80 for reach "R4", 60 for reach "R3" and "R2" and 20 for reach "R1". The value of "Muskingum X" is taken at 0.3 for all reach.

Reach	Initial Type	Initial Discharge (M3/S)	Muskingum K (HR)	Muskingum X	Number of Subreaches
R5	Discharge = Inflow		80	.5	1
R4	Discharge = Inflow		80	.5	1
R3	Discharge = Inflow		80	.5	1
R2	Discharge = Inflow		80	.5	1
R1	Discharge = Inflow		80	.5	1

Figure: Before calibration

Reach	Initial Type	Initial Discharge (M3/S)	Muskingum K (HR)	Muskingum X	Number of Subreaches
R5	Discharge = Inflow		120	0.3	1
R4	Discharge = Inflow		80	0.3	1
R3	Discharge = Inflow		60	0.3	1
R2	Discharge = Inflow		60	0.3	1
R1	Discharge = Inflow		20	0.3	1

Figure: After calibration

Baseflow method: Recession

The value of the recession constant is taken at 0.9 and the value of the ratio to peak is taken at 0.98 after calibration.

Subbasin	Initial Type	Initial Discharge (M3/S /KM2)	Initial Discharge (M3/S)	Recession Constant	Threshold Type	Threshold Flow (M3/S)	Ratio to Peak
Subbasin-1	Discharge		5	0.7	Ratio to Peak		0.6
Subbasin-2	Discharge		5	0.7	Ratio to Peak		0.6
Subbasin-3	Discharge		5	0.7	Ratio to Peak		0.6
Subbasin-4	Discharge		5	0.7	Ratio to Peak		0.6
Subbasin-5	Discharge		5	0.7	Ratio to Peak		0.6
S11	Discharge		5	0.7	Ratio to Peak		0.6

Figure: Before calibration

Subbasin	Initial Type	Initial Discharge (M3/S /KM2)	Initial Discharge (M3/S)	Recession Constant	Threshold Type	Threshold Flow (M3/S)	Ratio to Peak
Subbasin-1	Discharge		5	0.9	Ratio to Peak		0.98
Subbasin-2	Discharge		5	0.9	Ratio to Peak		0.98
Subbasin-3	Discharge		5	0.9	Ratio to Peak		0.98
Subbasin-4	Discharge		5	0.9	Ratio to Peak		0.98
Subbasin-5	Discharge		5	0.9	Ratio to Peak		0.98
S11	Discharge		5	0.9	Ratio to Peak		0.98

Figure: After calibration

Transform method: Synder's Unit Hydrograph

We have taken the value of peaking coefficient at 0.6 and the lag time is calculated using the following equation:

$t_l = CC_t\,(LL_c)^{0.3}$

Where:

L = length of the main stream from the outlet to the divide (mi/KM).

L_c = Length along the main stream to a point nearest the watershed centroid (mi/KM).

C_t= Timing coefficient (has been found to range from 1.8 to 2.2 for Appalachian Highland).

C=A conversion constant (0.75 for SI and 1.00 for foot-pound system).

For peak discharge, this method uses a peaking coefficient. peaking coefficient can range from 0.4

to 0.8, where larger values of C_p are associated with smaller values of C_t

Subbasin	Lag Time (HR)	Peaking Coefficient
Subbasin-1	13.7	0.6
Subbasin-2	13.9	0.6
Subbasin-3	11.6	0.6
Subbasin-4	17.7	0.6
Subbasin-5	18.8	0.6
S11	28.6	0.6

Values for Snyder's Unit Hydrograph

Loss method: SCS Curve Number

Here curve numbers are calculated using Land Use Land Cover map which is generated using ARC-GIS.

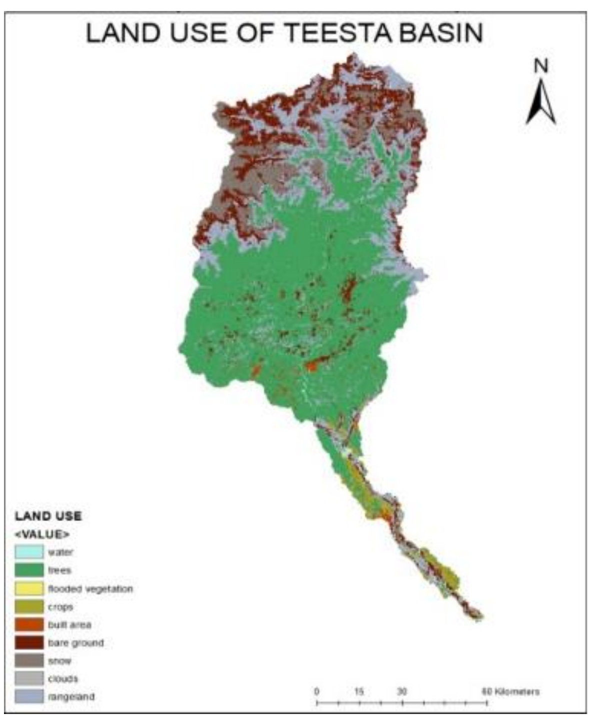
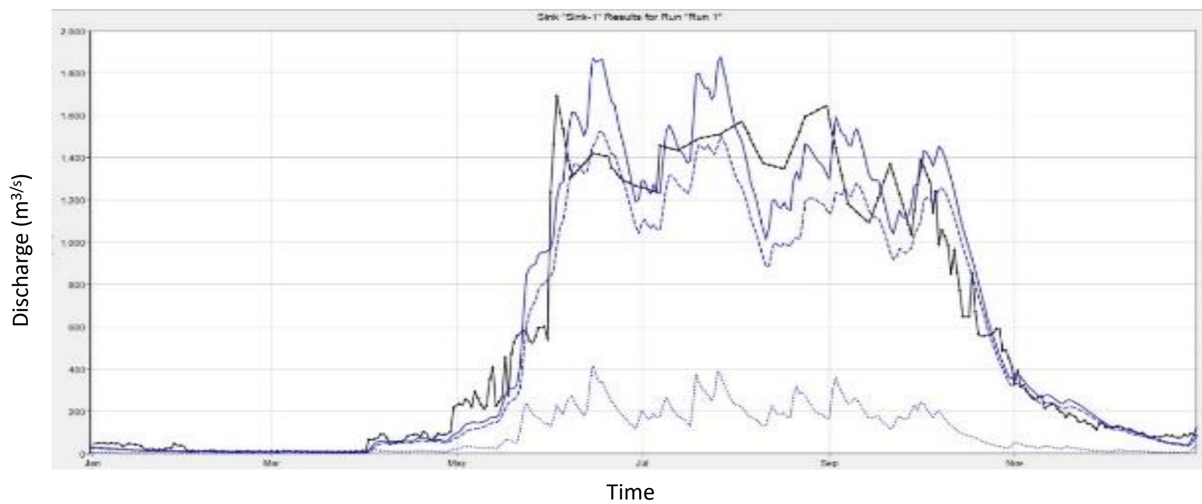


Figure: Land Use Map

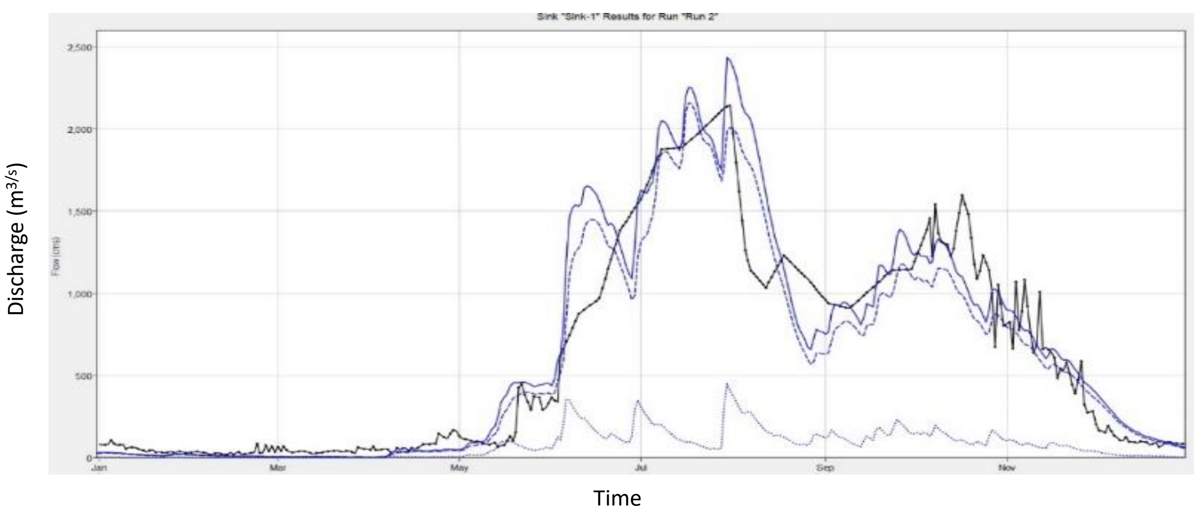
Subbasin	Initial Abstraction (MM)	Curve Number	Impervious (%)
Subbasin-1	30	77	20
Subbasin-2	30	69	27
Subbasin-3	30	65	27
Subbasin-4	30	61	31
Subbasin-5	30	61	31
S11	30	67	30

Figure: Value after Calibration

For validation we use discharge data of Dalia station of the year 2015. We also validate the model for the year 2016 for verifying purpose.



Validation for the year 2015



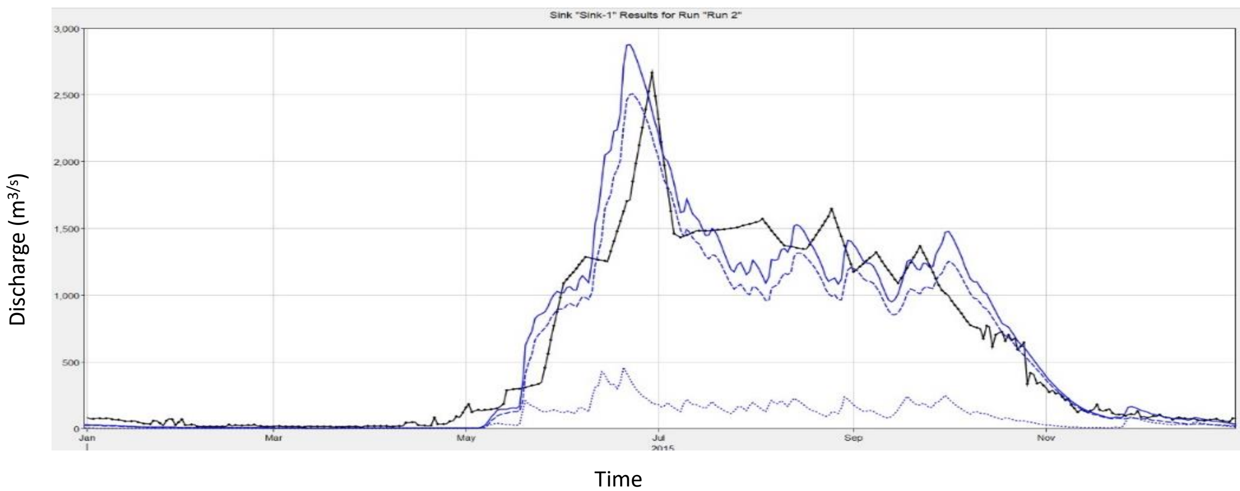
Validation for the year 2016

Result and Discussion:

In this study, our findings are the difference in peak discharge between simulated and observed flow, the difference in time of arrival of peak discharge, percent bias, RMSE standard deviation, and Nash-Sutcliffe values.

Difference in Discharge:

From the graph, it can be observed that there are minor differences in peak discharge value and the time of arrival of the peak discharge. So, it can be said that the simulated flow is representative of the observed flow. In the meantime, the time of arrival of the peak flow of simulated and observed flow is almost the same.



Simulated and observed discharge at Dalia station vs. Time

Calibrated Result:

From Table 2, in the year 2014, from calibration the percent bias was 5.16%, RMSE standard deviation was 0.4, and Nash-Sutcliffe was 0.874.

Observed Flow Gage Gage 1

Peak Discharge:	2664.8 (M3/S)	Date/Time of Peak Discharge:	29Jun2015, 00:00
Volume:	2037.74 (MM)		
RMSE Std Dev:	0.4	Nash-Sutcliffe:	0.874
Percent Bias:	5.16 %		

Table 2 *: Calibrated result:Year 2014*

Validated Result:

From Table 3, in the year 2015, from calibration the percent bias was 3.7%, RMSE standard deviation was 0.3, and Nash-Sutcliffe was 0.928.

Observed Flow Gage Gage 1

Peak Discharge:	1689.8 (M3/S)	Date/Time of Peak Discharge:	03Jun2015, 00:00
Volume:	2041.53 (MM)		
RMSE Std Dev:	0.3	Nash-Sutcliffe:	0.928
Percent Bias:	3.70 %		

Table 3: *Validated result: Year 2015*

From Table 4, in the year 2016, from calibration the percent bias was 5.71%, RMSE standard deviation was 0.4, and Nash-Sutcliffe was 0.868.

Observed Flow Gage Gage 1

Peak Discharge:	2147.4 (M3/S)	Date/Time of Peak Discharge:	31Jul2016, 00:00
Volume:	2146.06 (MM)		
RMSE Std Dev:	0.4	Nash-Sutcliffe:	0.868
Percent Bias:	5.71 %		

Table 4: *Validated result: Year 2016*

Discussion:

The model can be considered as a well-calibrated one based on the values of percent bias, RMSE standard deviation, and Nash-Sutcliffe. The calibrated model provided discharge values that were quite close to the observed discharge at station Dalia (SW291.5R). The arrival of peak discharge from the model and from observed data was close enough that the model can be considered as a forecasting tool. The Teesta River model has been run based on rainfall and observed data that can generate an effect on water flow at the station located in Dalia (SW291.5R). From this model, the hydrograph, peak discharge water volume, and peak discharge time have been generated. The simulated model can be used as a future flood forecasting tool as the calibrated model shows when the peak discharge will come. The model can be useful for taking flood mitigation measures by incorporating projected precipitation of the future.

Conclusion:

The lower Teesta basin has been examined during this work, and a simulation model of it has been built. The flood simulation model was performed based on the last rainfall event in this location and some information on water flow, discharge, and peak discharge time of chosen places was created, which is highly useful for flood analysis. Based on any significant rainfall, the peak discharge timing and volume may be anticipated in the actual world, and the entire area of flooding can be calculated. As a result, residents in this area may be concerned before the water arrives. The overall loss of life and property can be reduced by this method. This study employs highly scientific and cutting-edge approaches, as well as dependable satellite data products. As a result, this is a one-of-a-kind endeavor in this region, demonstrating the manner of flood forecasting based on superior technical know-how. When the local government implements the approach, the region's oppressed aborigines will profit the most. This concept can also be used in other flood-prone areas across the world for early warning and rescue operations.